DATA 606 Data Project Proposal

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Data Preparation

```
# load data
library(tidyverse)
library(scales)

url <- "https://raw.githubusercontent.com/thomasthinks/college-roi-data/refs/heads/main/best-value-colle
df <- read_csv(url)
df <- df |>
  mutate(control = factor(control,
                          levels = c("public", "private-nonprofit", "private-forprofit")))
```

```
head(df)
```

```
## # A tibble: 6 x 7
##   state state_name rank_in_state institution city  control npv_30yr_resident_usd
##   <chr> <chr>              <dbl> <chr>       <chr> <fct>                   <dbl>
## 1 AL    Alabama                1 Columbia S~ Oran~ privat~               4210121
## 2 AL    Alabama                2 University~ Hunt~ public                2488746
## 3 AL    Alabama                3 Auburn Uni~ Aubu~ public                2200730
## 4 AL    Alabama                4 The Univer~ Tusc~ public                1820478
## 5 AL    Alabama                5 Tuskegee U~ Tusk~ privat~               1809626
## 6 AL    Alabama                6 Faulkner U~ Mont~ privat~               1441726
```

Research question

Of the top-ranking colleges from each state in the United States, is there a relationship between the type of institution, whether public, private non-profit or private profit, and the 30-year net present value (NPV) of attendance in the state?

The results obtained are only from top-ranking institutions since the data only gives details of the top-ranking colleges in each state.

Cases

What are the cases, and how many are there?

Each case is one institution listed in a state's top-value ranking (up to 15 per state). There are 607 cases across 50 states.

Data collection

The NPV figures are calculated, notobserved. The data maintainers, LE TEEN, computed them from three public sources:

- FREOPP higher-education ROI data (2021), built on College Scorecard earnings cohorts

- NCES IPEDS 2023-24 - tuition, room and board
- BEA Regional Price Parities - 2023, cost-of-living adjustment

The NPV here refers to the value of education today. It reflects the net amount the student is expected to earn in 30 years from his/her education after deducting the cost of earning the degree and considering the time value of future dollars (at a discount of 3% per annum after adjusting for inflation). The schools are ranked according to their NPVs in each state.

Type of study

The NPV calculations are produced by observing what has actually happen. Students choose where to go to college so we have recorded their actual earnings, tuition and costs.

The NPVs have been calculated using actual data. We have taken actual graduate earnings (College Scorecard), tuition and housing costs (IPEDS) and cost of living differences (BEA) and used a formula on them to produce the NPVs. This means it has been transformed.

Data Source

LE TEEN (2026). *College ROI Data: 30-year net present value of U.S. colleges and majors.* Zenodo. https://doi.org/10.5281/zenodo.21351602

Repository: https://github.com/thomasthinks/college-roi-data (CC BY 4.0). File used: best-value-colleges-by-state.csv

Describe your variables?

- Dependent variable: npv_30yr_resident_usd, the 30-year net present value, in U.S. dollars, of attending an institution as an in-state resident. This is a quantitative variable (continuous). It is the outcome I am trying to explain, so it is the dependent variable in the model.
- Independent variable: control, the type of institution (public, private-nonprofit, or private-forprofit). This is a qualitative variable (categorical, with three levels).
- Other**:** state (qualitative), rank_in_state (quantitative, 1 to 15), institution and city (identifiers).

Because the response is quantitative and the explanatory variable is categorical with three groups, the planned analysis is a comparison of mean NPV across the three groups.

Relevant summary statistics

```
glimpse(df)
```

```
## Rows: 607
## Columns: 7
## $ state                <chr> "AL", "AL", "AL", "AL", "AL", "AL", "AL", "AL", ~
## $ state_name           <chr> "Alabama", "Alabama", "Alabama", "Alabama", "Ala~
## $ rank_in_state        <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 1~
## $ institution          <chr> "Columbia Southern University", "University of A~
## $ city                 <chr> "Orange Beach", "Huntsville", "Auburn", "Tuscalo~
## $ control              <fct> private-forprofit, public, public, public, priva~
## $ npv_30yr_resident_usd <dbl> 4210121, 2488746, 2200730, 1820478, 1809626, 144~
```

```
sum(is.na(df))
```

```
## [1] 0
```

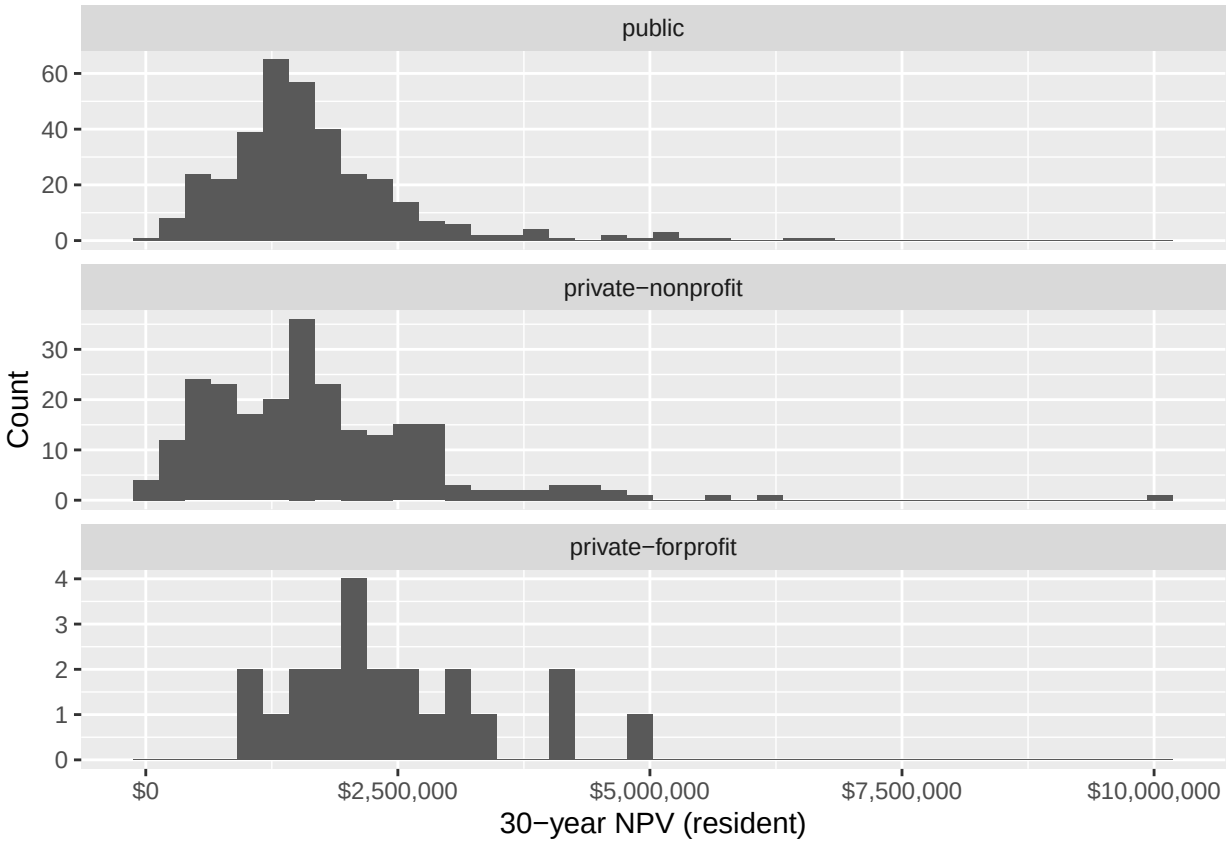
```r
df |>
  group_by(control) |>
  summarise(n = n())
```

```
## # A tibble: 3 x 2
##   control               n
##   <fct>             <int>
## 1 public              348
## 2 private-nonprofit   237
## 3 private-forprofit    22
```

```r
df |>
  group_by(control) |>
  summarise(n = n(),
            mean = mean(npv_30yr_resident_usd),
            median = median(npv_30yr_resident_usd),
            sd = sd(npv_30yr_resident_usd),
            min = min(npv_30yr_resident_usd),
            max = max(npv_30yr_resident_usd)) |>
  mutate()
```

```
## # A tibble: 3 x 7
##   control               n     mean  median       sd    min      max
##   <fct>             <int>    <dbl>   <dbl>    <dbl>  <dbl>    <dbl>
## 1 public              348 1663965. 1480576  958868.   4837  6593271
## 2 private-nonprofit   237 1717857. 1572337 1200038.  28600 10062539
## 3 private-forprofit    22 2446360. 2214730 1018072. 979623  4868871
```

```r
ggplot(df, aes(npv_30yr_resident_usd)) +
  geom_histogram(bins = 40) +
  facet_wrap(~ control, scales = "free_y", ncol = 1) +
  scale_x_continuous(labels = dollar) +
  labs(x = "30-year NPV (resident)", y = "Count")
```



The distributions are right-skewed, so medians are more representative than means. The private for-profit group is small with a count of only 22, so comparisons involving it will be imprecise.